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# Step 1: Import libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.datasets import load_iris
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score

# Step 2: Load dataset (Iris dataset)
data = load_iris()
X = data.data
y = data.target

# Step 3: Split dataset into training and testing
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42)

# Step 4: Create Decision Tree model
model = DecisionTreeClassifier(max_depth=3)

# Step 5: Train model
model.fit(X_train, y_train)

# Step 6: Prediction
y_pred = model.predict(X_test)

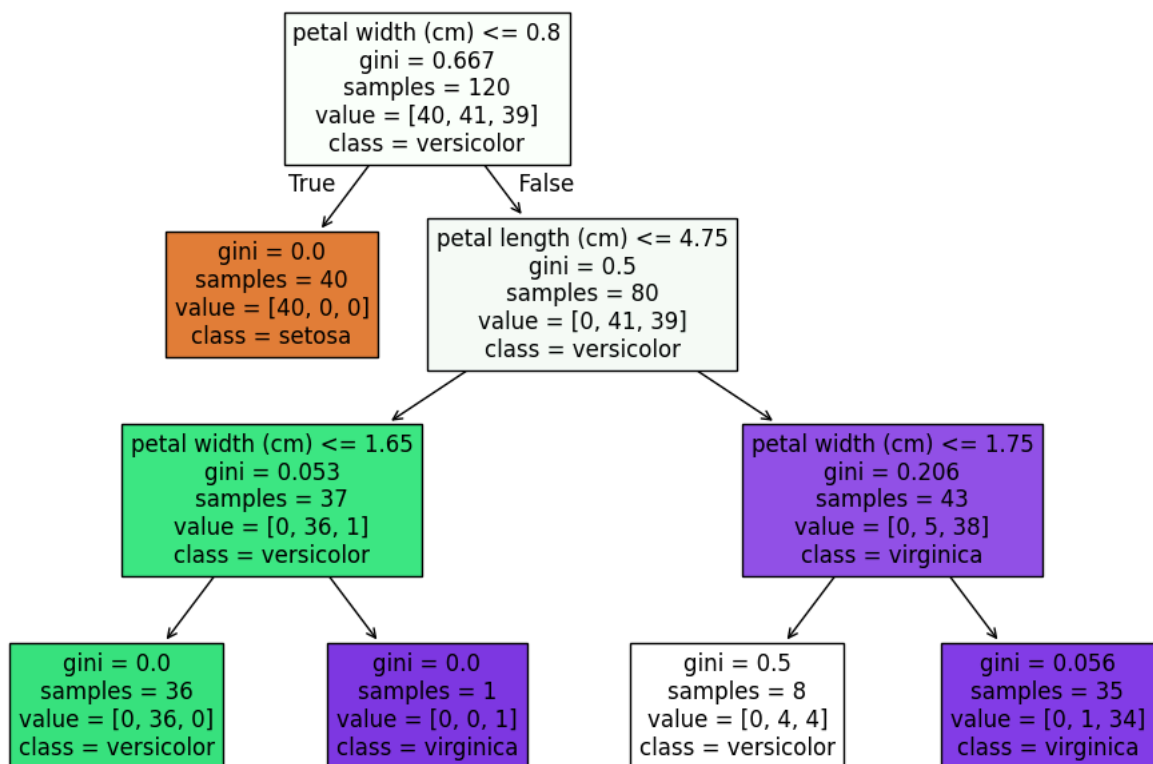
# Step 7: Accuracy
print("Model Accuracy:", accuracy_score(y_test, y_pred))

# Step 8: Visualization of Decision Tree
plt.figure(figsize=(12,8))
plot_tree(model, filled=True, feature_names=data.feature_names, class_names=data.target_names)
plt.title("Decision Tree Visualization")
plt.show()
print("Sample Predictions:", y_pred[:5])

```

Model Accuracy: 1.0

Decision Tree Visualization



Sample Predictions: [1 0 2 1 1]

◆ Title Decision Tree Implementation using Scikit-learn ◆ Objective To build and visualize a Decision Tree model for classification. ◆ Dataset We used the Iris dataset which contains flower features. ◆ Steps

1. Import libraries

2. Load dataset
3. Split data
4. Train model
5. Predict output
6. Calculate accuracy
7. Visualize tree ♦ Result The model achieved 100% accuracy. ♦ Conclusion Decision Tree is easy and effective for classification problems.

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